Validity Analysis: MILU

Target context: Mymensingh Environmental Scientist — Cross-Regional South Asian Agricultural Knowledge

Overall risk: **HIGH**

Deployment Context

Use case: Task: Environment science and geograpy-specific agricultural knowledge understanding

LLMs evaluated: Generic frontier model vs small llms as well as region specific llms

Domain: Environmental science

Setting: Environment science research utility assessment using llm and local-context

Target population: An environment scientist from Mymensingh, Bangladesh wants to get knowledge about different south asian region specific agricultural practice. So having different region and language specific agricultural and demographic context understanding (e.g. telegu spoken area, hydrabad India and agricultural practice/problems vs farmers practice/problems from Mymensingh, Bangladesh~bengali spoken in Mymensingh accented dialect.)

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology ⚠	2	Significant gaps	high
Input Content ⚠	1	Serious concern	high
Input Form ✓	3	Moderate gaps	high
Output Ontology	2	Significant gaps	high
Output Content ⚠	2	Significant gaps	high
Output Form ✓	3	Moderate gaps	high
Average	**2.2**		

⚠ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

MILU is fundamentally misaligned with the Mymensingh environmental-scientist deployment context. The benchmark is by design India-centric across all six validity dimensions: its 41-subject taxonomy lacks Bangladesh-specific agro-ecological subdomains; its content originates exclusively from Indian competitive exams with no Bangladeshi farming, dialect, or trans-border hydrology representation; its Bengali corpus reflects West Bengal exam register and is heavily machine-translated (100% of the sampled validation Bengali split); its annotation pool excluded Bangladeshi subject-matter experts; and its single-correct-answer MCQ format cannot accommodate the cross-border knowledge pluralism that trans-border water-policy questions demand. Dataset analysis empirically confirms these structural gaps: Bangladesh appears in 215 sampled examples only as a wrong-answer distractor, Agriculture-labeled questions concern Indian schemes and corporate trivia rather than agronomy, and Telangana/AP agro-ecological content is absent. MILU's strengths — broad Indic-script coverage, consistent MCQ structure, broad domain diagnostic breadth — make it useful as a generic Indic-LLM screening tool, but it provides essentially no discriminative signal for Bangladeshi agricultural and environmental science knowledge.

Practical Guidance

What This Benchmark Measures

MILU measures Indian competitive-exam general knowledge across 11 Indic languages and 8 broad domains. For the Mymensingh deployment, it can usefully measure (a) whether candidate LLMs render Bengali and Telugu scripts correctly (input_form strength) and (b) whether they have basic STEM and general-knowledge capability across Indic languages (output_form strength enables cross-model comparison). It cannot measure agricultural science depth, Bangladeshi agro-ecological knowledge, trans-border policy reasoning, or Bangladeshi Bengali register competence — the core constructs the deployment cares about.

Construct Depth

Construct depth is shallow for the target deployment. Even within the relevant Environmental Sciences domain, sampled content is geography trivia and Indian-policy-scheme recall, not agronomic, soil-science, or wetland-ecology reasoning (input_ontology score 2, input_content score 1). The Agriculture subject label exists but maps to institutional and corporate trivia. MILU provides surface-level breadth without the technical depth a professional environmental scientist would need to assess LLM reliability for their query workload.

What Else You Need

Significant supplementation is required: (a) a Bangladesh-specific agricultural knowledge benchmark covering haor/beel/char-land systems and boro/aman cycles (none currently exists per GAP-1, GAP-8); (b) BEnQA [WEB-6] as a Bangladeshi Bengali register baseline to disentangle language-competence from cultural-knowledge gaps; (c) a trans-border policy benchmark with pluralistic ground-truth structure to address output_ontology and output_content gaps; (d) Telangana/AP sub-regional agro-ecological items if cross-reference zones are core to the use case; and (e) generation-mode and CoT evaluation to address [Q85] log-likelihood limitations for reasoning-heavy queries.

ID	Evidence
Q85	“Finally, our evaluation primarily relies on the log-likelihood approach, which may yield different results compared to other established methods, such as generation-based evaluation and chain-of-thought (CoT) prompting.” (p.9)
WEB-6	BEnQA exists as a separate Bangladeshi Bengali science benchmark, but is not integrated into MILU (https://arxiv.org/abs/2403.10900)